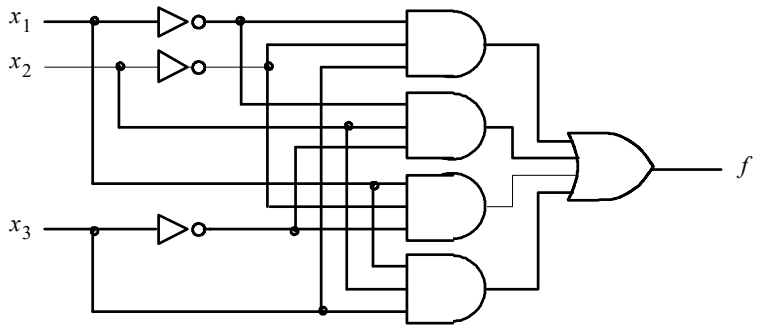
Department of Electrical and Computer Engineering
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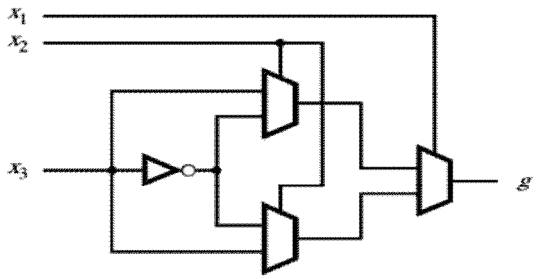
ELEC-271 Digital Systems

Problem Set 2

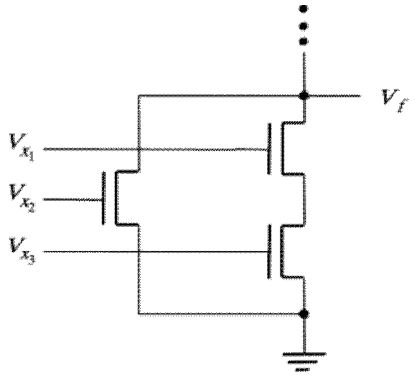
1. Consider the circuit shown below. Show the truth table for the logic function *f*. If each gate in the circuit is implemented as a CMOS gate, how many transistors are needed?



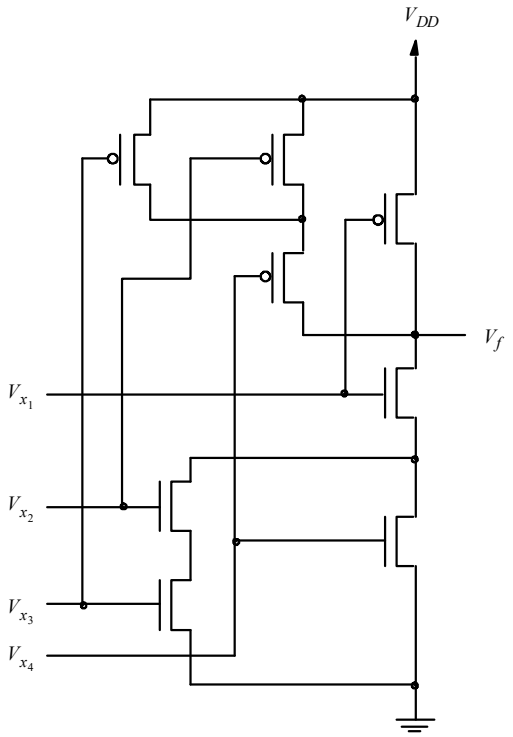
2. Show that the following circuit is functionally equivalent to the circuit in question 1.



3. The following circuit shows half of a CMOS circuit. Derive the other half that contains PMOS transistors.



4. Give the truth table for the CMOS circuit below. Derive a canonical SOP expression for the truth table.



5. Derive a CMOS complex gate for the minimum SOP expression for the logic function $f(x_1, x_2, x_3, x_4) = \sum m(0,1,2,4,6,8,10,12,14)$.

6. Using the style of drawing in Figure 3.66 of the textbook, draw a picture of a PLA programmed to implement $f(x_1, x_2, x_3) = \sum m(1,2,4,7)$. The PLA should have the inputs x_1, x_2, and x_3; the product terms P_1, P_2, P_3, and P_4; and the outputs f_1 and f_2.

7. Consider the function $f(x_1, x_2, x_3) = x_1\bar{x}_2 + x_1x_3 + x_2\bar{x}_3$. Show a circuit using 5 two-input lookup-tables (LUTs) to implement this expression. Give the truth table implemented in each LUT. You do not need to show the wires in the FPGA.

8. Assume that a gate array contains the type of logic cell depicted in the following figure. The inputs in_1, in_2, …, in_7 can be connected to either 1, 0, or any logic signal. Show how the logic cell can be used to realize $f = x_1x_3 + x_2x_3$.

